

# Update and Delete Documents

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MongoDB provides powerful methods to **update** or **remove** documents from a collection.

This section covers:

- `updateOne()` , `updateMany()` , `$set` , `$inc`
- `deleteOne()` , `deleteMany()`
- `replaceOne()`
- Understanding `ObjectId`

**Note:** All code examples are for MongoDB Compass shell (mongosh).

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## Updating Documents

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### updateOne

Updates the **first** document that matches the filter.

```
// Change Ali's course to Advanced MongoDB
db.students.updateOne(
  { name: 'Ali' },
  { $set: { course: 'Advanced MongoDB' } }
)
```

### updateMany

Updates **all** documents that match the filter.

```
// Add 1 year to age of all enrolled students
db.students.updateMany(
```

```

    { enrolled: true },
    { $inc: { age: 1 } }
  )

// Update all MongoDB students to have a new teacher
db.students.updateMany(
  { course: 'MongoDB' },
  { $set: { teacherId: ObjectId("507f1f77bcf86cd799439011") } }
)

```

## Common Update Operators

- `$set` : Sets the value of a field
- `$unset` : Removes a field
- `$inc` : Increments a numeric field
- `$push` : Adds an item to an array
- `$pull` : Removes an item from an array
- `$addToSet` : Adds unique items to an array

Examples:

```

// Add a new grade to Sara's grades array
db.students.updateOne(
  { name: 'Sara' },
  { $push: { grades: 96 } }
)

// Remove enrolled field from Ravi
db.students.updateOne(
  { name: 'Ravi' },
  { $unset: { enrolled: "" } }
)

// Increment teacher's experience by 1
db.teachers.updateOne(
  { name: 'Dr. Kumar' },

```

```
{ $inc: { experience: 1 } }  
)
```

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## Replacing a Document

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### replaceOne

Replaces the entire document with a new one (except the `_id`).

```
db.students.replaceOne(  
  { name: 'Ravi' },  
  {  
    name: 'Ravi Kumar',  
    age: 24,  
    course: 'Python',  
    enrolled: true,  
    teacherId: ObjectId("507f1f77bcf86cd799439012"),  
    grades: [80, 85, 82],  
    email: 'ravi.kumar@school.edu'  
  }  
)
```

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## Deleting Documents

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### deleteOne

Deletes the **first** matching document.

```
// Delete a specific student  
db.students.deleteOne({ name: 'Kabir' })
```

## deleteMany

Deletes all matching documents.

```
// Delete all students who are not enrolled
db.students.deleteMany({ enrolled: false })

// Delete all students with low average grades
db.students.deleteMany({
  $expr: {
    $lt: [{ $avg: "$grades" }, 75]
  }
})
```

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## Working with ObjectId

Each document in MongoDB has a unique `_id` field of type `ObjectId`.

In MongoDB Compass, ObjectId is available globally:

```
// Find a student by _id
db.students.findOne({
  _id: ObjectId("64bd2e183dd4e6402f10388f")
})

// Update a teacher by _id
db.teachers.updateOne(
  { _id: ObjectId("507f1f77bcf86cd799439011") },
  { $set: { office: "Room 301" } }
)
```

# Practical Examples

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## Update Student's Teacher

```
// Move all Express students to a different teacher
db.students.updateMany(
  { course: 'Express' },
  { $set: {
    teacherId: ObjectId("507f1f77bcf86cd799439012"),
    course: 'Advanced Express'
  }}
)
```

## Bulk Grade Update

```
// Add bonus points to all students of a specific teacher
db.students.updateMany(
  { teacherId: ObjectId("507f1f77bcf86cd799439011") },
  { $push: { grades: 5 } } // Add 5 bonus points
)
```

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## Summary

- Use `updateOne()` or `updateMany()` with operators like `$set`, `$inc`, `$push`
- Use `replaceOne()` to completely replace documents
- Use `deleteOne()` or `deleteMany()` to remove documents
- `ObjectId` is globally available in MongoDB Compass
- Always verify your filters before running update/delete operations